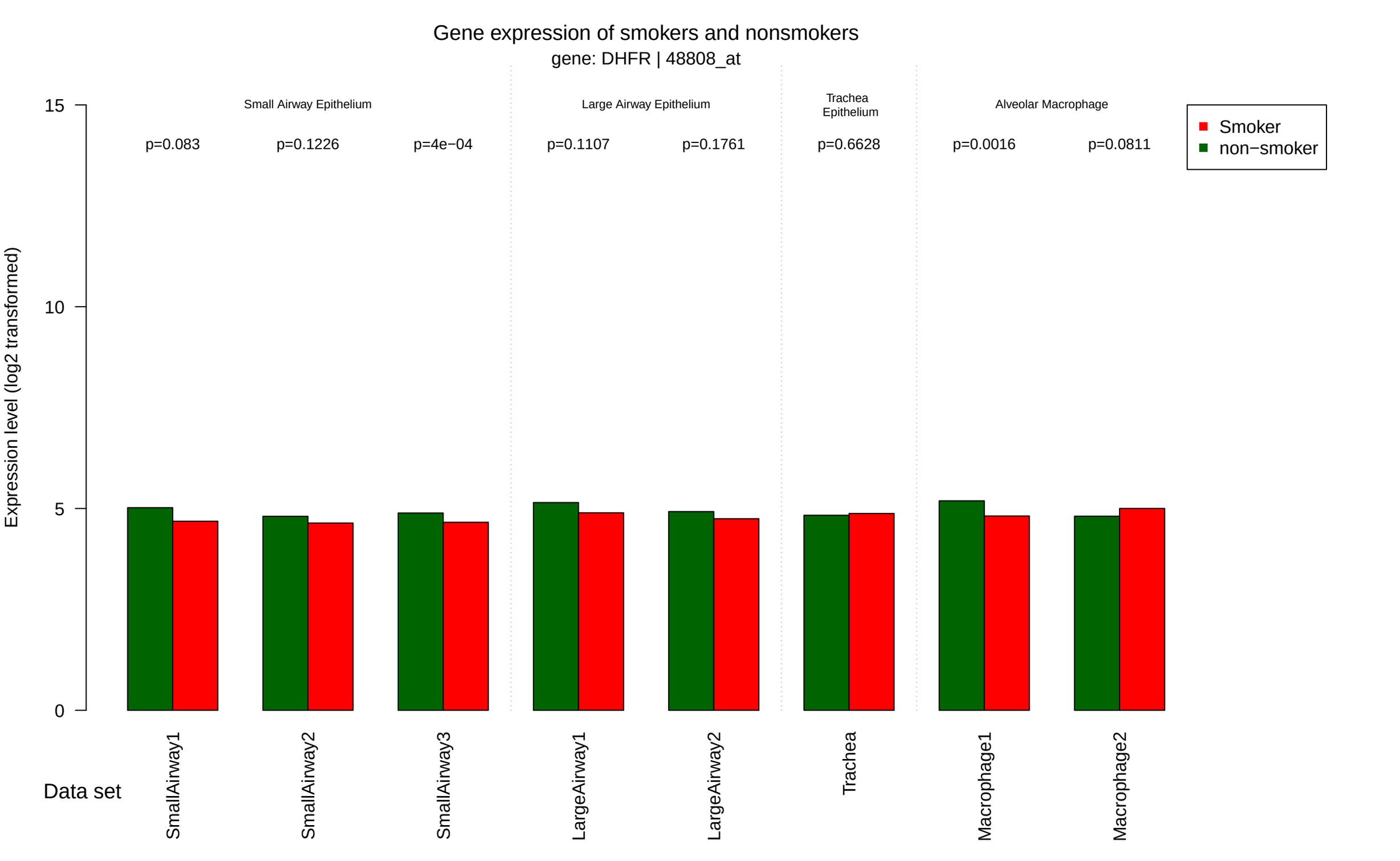
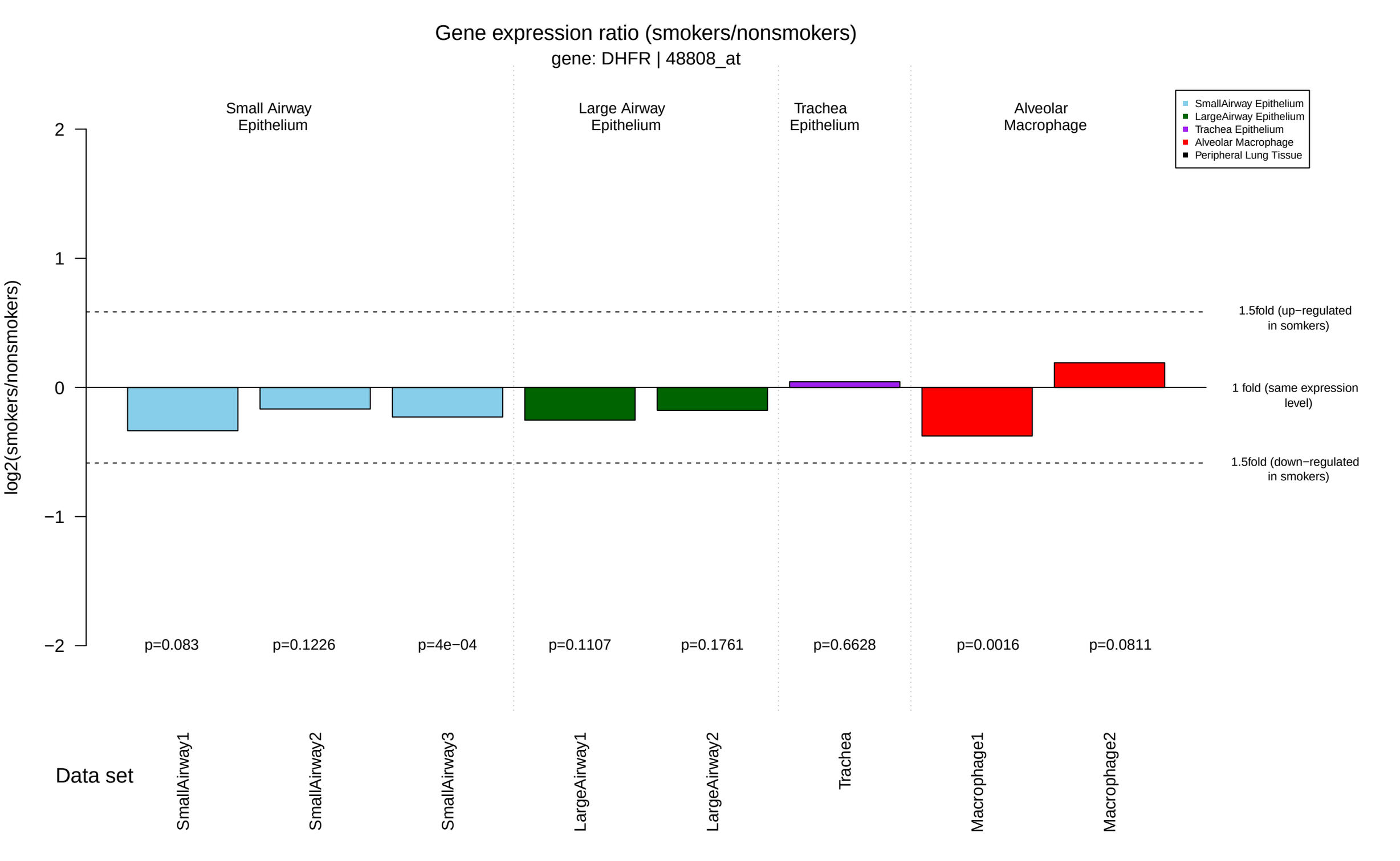
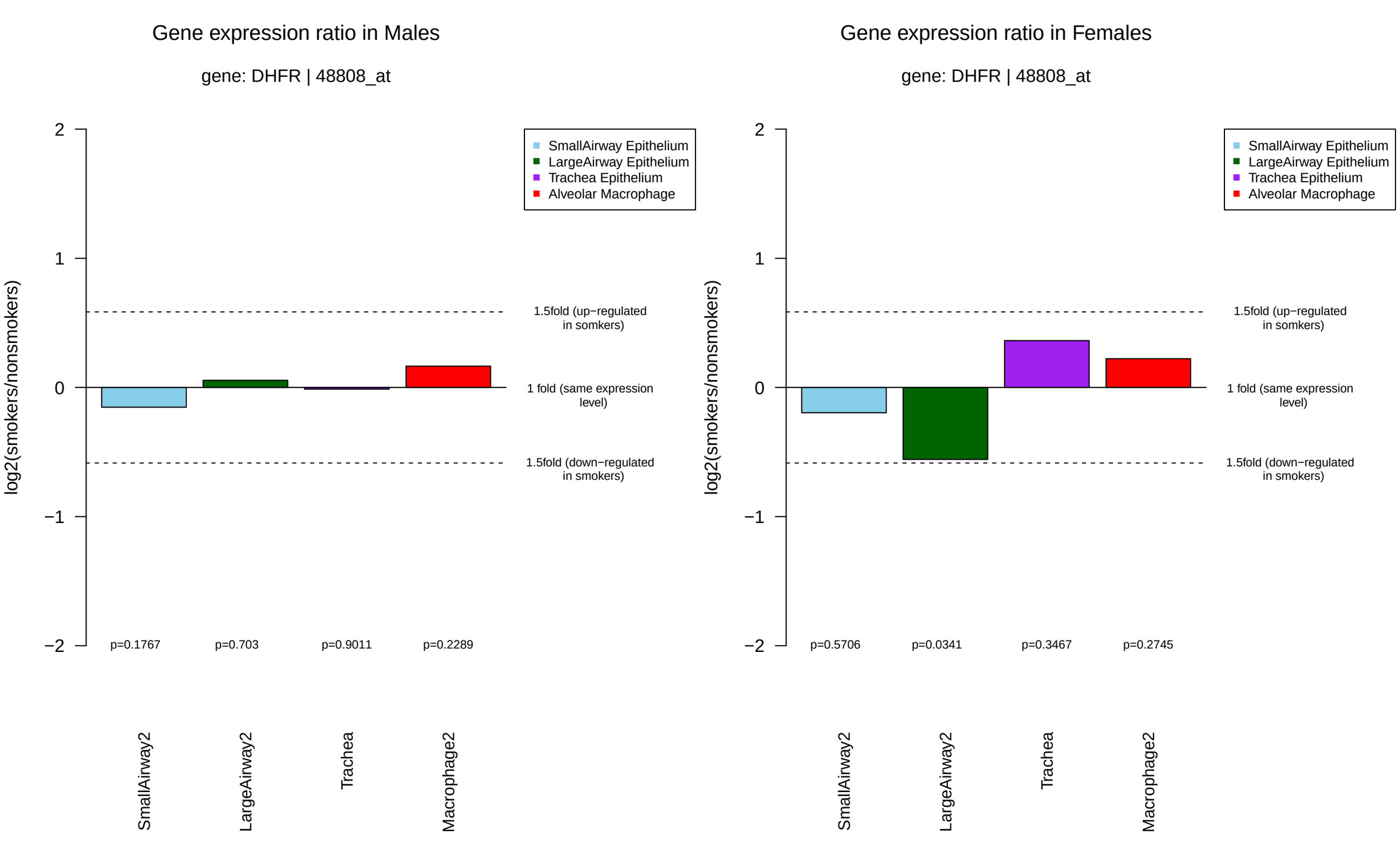
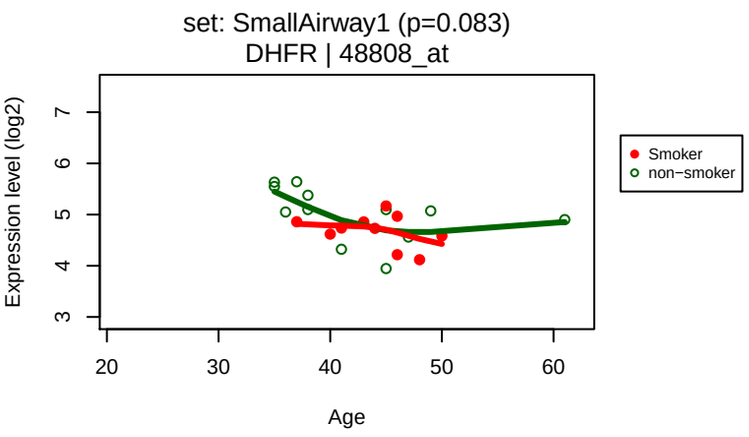
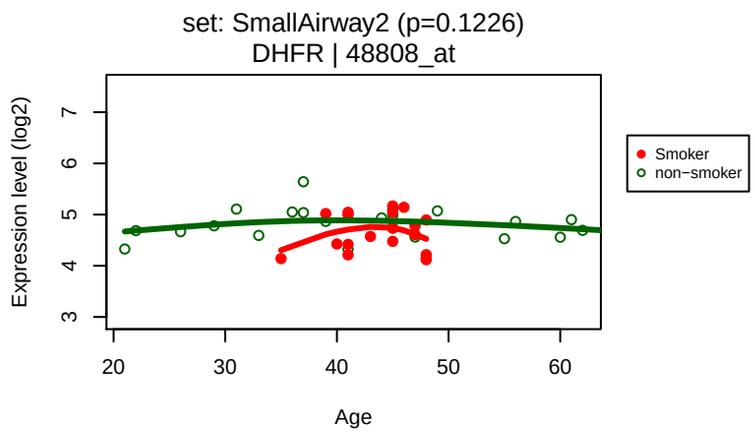
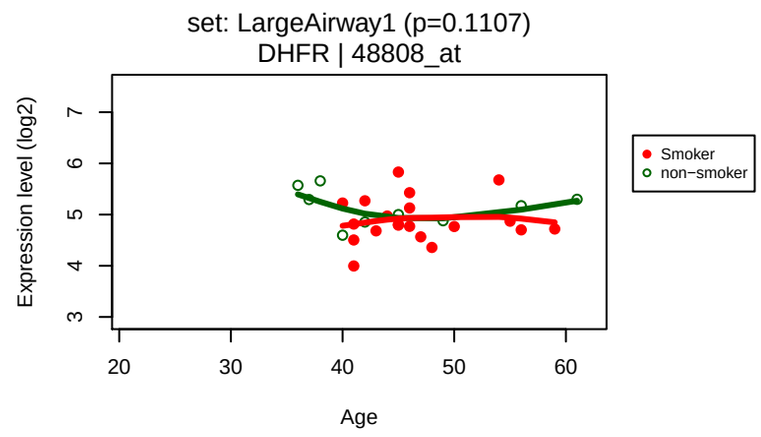

Gene expression of smokers and nonsmokers
gene: DHFR | 48808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.083
p=0.1226
p=4e−04
p=0.1107
p=0.1761
p=0.6628
p=0.0016
p=0.0811
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

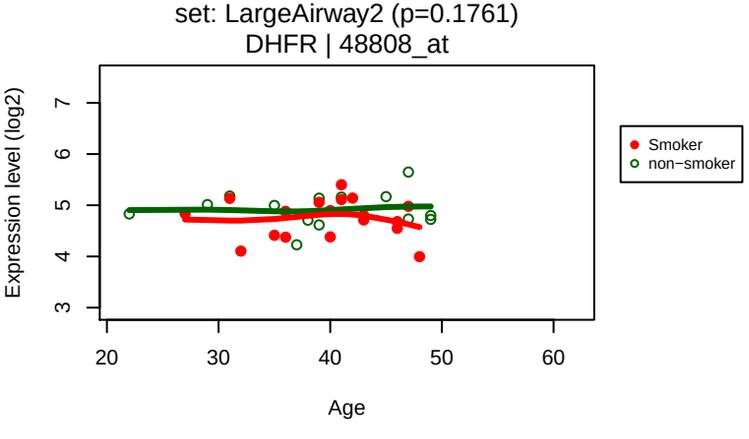

Gene expression ratio (smokers/nonsmokers)
gene: DHFR | 48808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.083
p=0.1226
p=4e−04
p=0.1107
p=0.1761
p=0.6628
p=0.0016
p=0.0811
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

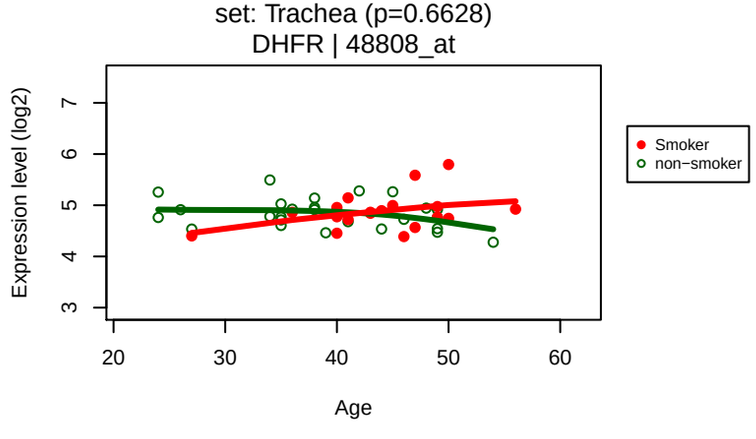

Gene expression ratio in Males
gene: DHFR | 48808_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1767
p=0.703
p=0.9011
p=0.2289
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DHFR | 48808_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5706
p=0.0341
p=0.3467
p=0.2745
SmallAirway2
LargeAirway2
Trachea
Macrophage2

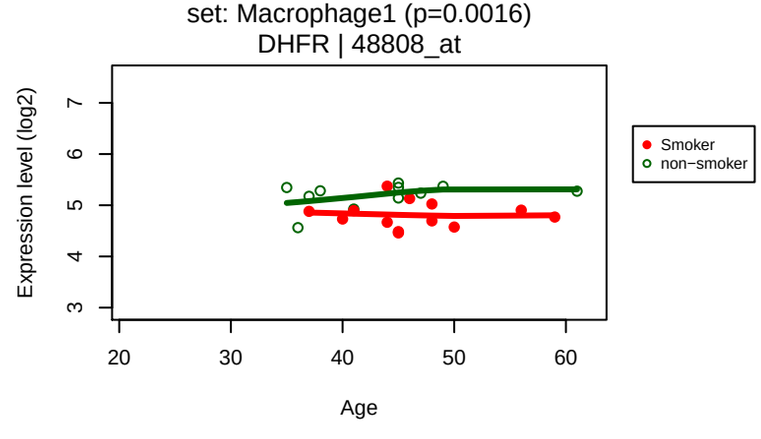

set: SmallAirway1 (p=0.083)
DHFR | 48808_at
Expression level (log2)
Age
Smoker
non−smoker

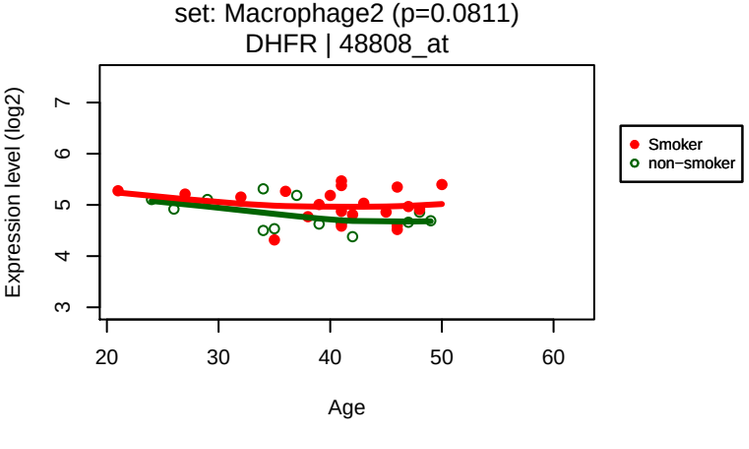

set: SmallAirway2 (p=0.1226)
DHFR | 48808_at
Expression level (log2)
Age
Smoker
non−smoker

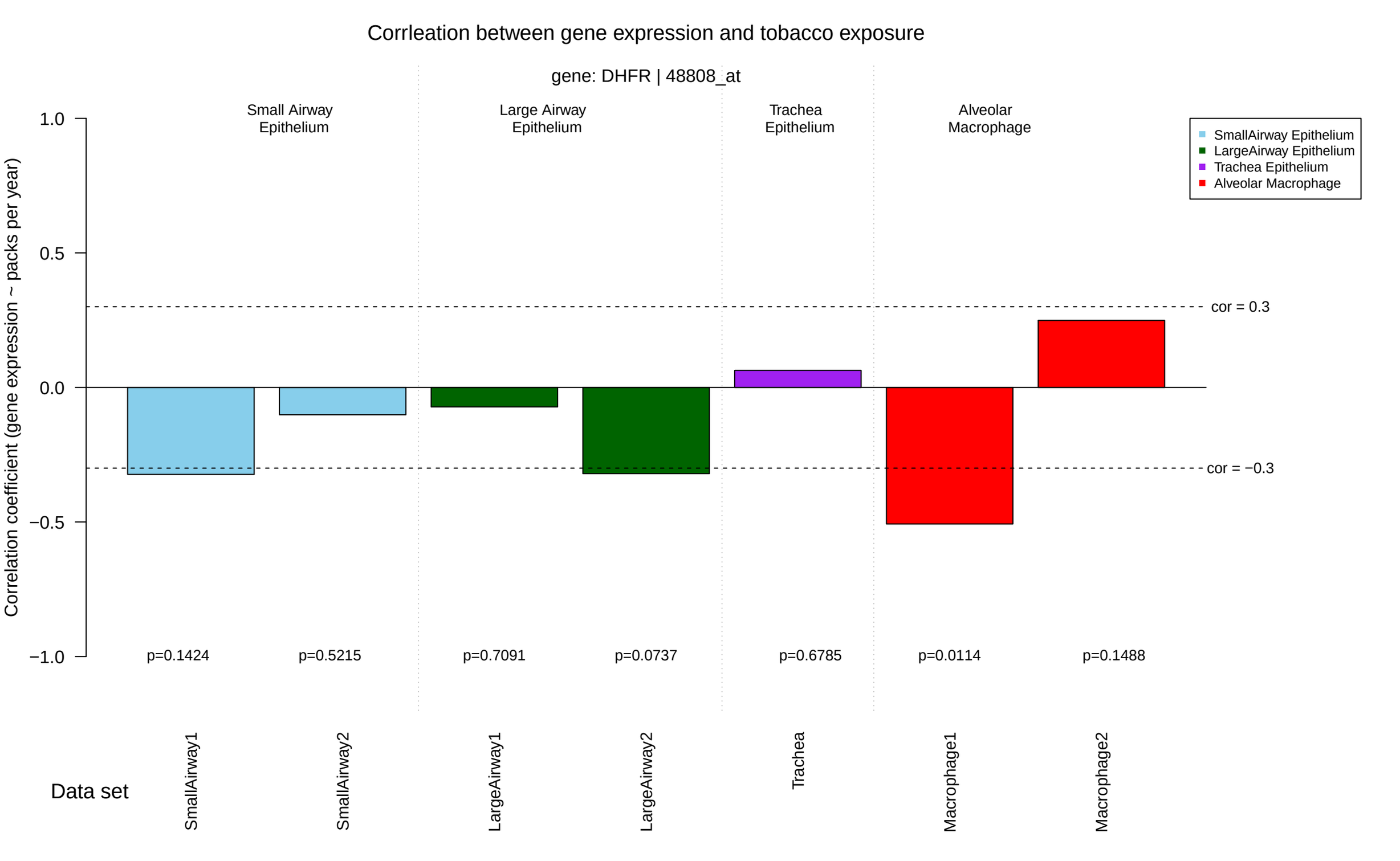

set: LargeAirway1 (p=0.1107)
DHFR | 48808_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1761)
DHFR | 48808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6628)
DHFR | 48808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0016)
DHFR | 48808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0811)
DHFR | 48808_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DHFR | 48808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1424
p=0.5215
p=0.7091
p=0.0737
p=0.6785
p=0.0114
p=0.1488
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2